

5 Lecture 5: Fourier Transform and PDEs

Fourier transform and PDEs [see Evans, §4.3.1].

5.1 From Fourier series to the Fourier transform

For a periodic function $f: [-\pi, \pi] \rightarrow \mathbb{R}$, recall the complex Fourier-series representation

$$f(x) = \frac{1}{\sqrt{2\pi}} \sum_{k \in \mathbb{Z}} a_k e^{ikx}, \quad (5.1)$$

where

$$a_k = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} e^{-iky} f(y) dy. \quad (5.2)$$

Here k is the **frequency** of the periodic oscillation $x \mapsto e^{ikx}$, and a_k is the “**amplitude**” of f at frequency k .

Under suitable hypotheses, a periodic function can be represented as an infinite superposition of oscillations with integer frequencies.

Fourier transform.

It generalizes Fourier series to cover all real-valued frequencies.

Recall L^p spaces. For $1 \leq p < \infty$,

$$\|f\|_p = \|f\|_{L^p} := \left(\int_{\mathbb{R}^n} |f(x)|^p dx \right)^{1/p}, \quad (5.3)$$

and

$$\|f\|_{\infty} = \|f\|_{L^{\infty}} := \text{ess sup } |f|. \quad (5.4)$$

Thus $f \in L^p$ means $\|f\|_p < \infty$. The same definitions apply to complex-valued functions $f: \mathbb{R}^n \rightarrow \mathbb{C}$.

Definition 5.1 (Fourier transform). For $f \in L^1(\mathbb{R}^n)$, define

$$\mathcal{F}(f)(\xi) = \widehat{f}(\xi) := (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-i\xi \cdot x} f(x) dx, \quad i = \sqrt{-1}. \quad (5.5)$$

Here x is the “physical variable” and ξ is the “frequency variable.” The source interprets $\widehat{f}(\xi)$ as the “**amplitude**” of f at frequency ξ .

Equivalently,

$$\widehat{f}(\xi) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \cdots \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-i(\xi_1 x_1 + \cdots + \xi_n x_n)} f(x_1, \dots, x_n) dx_1 \cdots dx_n. \quad (5.6)$$

Definition 5.2 (Inverse Fourier transform). For $g \in L^1(\mathbb{R}^n)$, define

$$\mathcal{F}^{-1}(g)(x) = \check{g}(x) := (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{i\xi \cdot x} g(\xi) d\xi. \quad (5.7)$$

5.2 Why is the Fourier transform interesting?

A non-rigorous reasoning. Every “reasonable” function can be written as an integral of oscillations with frequencies ξ :

$$f(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} \widehat{f}(\xi) e^{i\xi \cdot x} d\xi. \quad (5.8)$$

That is,

$$f = (\widehat{f})^\vee. \quad (5.9)$$

We will prove this later.

For PDEs, formally differentiating (5.8) gives

$$\frac{\partial f}{\partial x_j}(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} i\xi_j \widehat{f}(\xi) e^{i\xi \cdot x} d\xi. \quad (5.10)$$

Thus

$$\widehat{\frac{\partial f}{\partial x_j}}(\xi) = i\xi_j \widehat{f}(\xi). \quad (5.11)$$

So, under the Fourier transform, differentiation becomes multiplication. **By Fourier transform, we can reduce a PDE to an algebraic equation or to a PDE with fewer derivatives, hence to a simpler equation.**

Exercise 5.3 (Fourier transform identities). Show, at least formally, for $f(x), g(x), \varphi(\xi), \psi(\xi)$ (the source assumes these functions lie in $L^1 \cap L^2$), that

$$\widehat{f * g}(\xi) = (2\pi)^{n/2} \widehat{f}(\xi) \widehat{g}(\xi), \quad (5.12)$$

$$\widehat{fg}(\xi) = (2\pi)^{-n/2} (\widehat{f} * \widehat{g})(\xi), \quad (5.13)$$

$$(\varphi * \psi)^\vee(x) = (2\pi)^{n/2} \varphi^\vee(x) \psi^\vee(x), \quad (5.14)$$

$$(\varphi\psi)^\vee(x) = (2\pi)^{-n/2} (\varphi^\vee * \psi^\vee)(x). \quad (5.15)$$

Hint: do (5.12) and (5.14) first, then use $(\widehat{f})^\vee = f$ and $(\varphi^\vee)^\vee = \varphi$ to obtain the other two. For $f, g \in L^1(\mathbb{R}^n)$, the computation for (5.12) is rigorous by Fubini’s theorem.

5.3 Example: the heat equation in $\mathbb{R}^n$

Example. Heat equation in $\mathbb{R}^n$ (cf. Evans, §2.3.1). Consider the **initial value problem (IVP)**

$$\begin{cases} \frac{\partial u}{\partial t} - \Delta u = 0, & (x, t) \in \mathbb{R}^n \times \mathbb{R}_+, \\ u(x, 0) = u_0(x), & x \in \mathbb{R}^n. \end{cases} \quad (5.16)$$

Here u_0 is the **initial value**.

Take the Fourier transform in the x -variable:

$$\widehat{u}(\xi, t) := (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-i\xi \cdot x} u(x, t) dx. \quad (5.17)$$

Formally, the transform gives the correspondence

physical side	Fourier side
$u_0(x)$	$\widehat{u}_0(\xi)$
$u(x, t)$	$\widehat{u}(\xi, t)$
u_t	$\partial_t \widehat{u}$
u_{x_j}	$i\xi_j \widehat{u}$
$u_{x_j x_j}$	$-\xi_j^2 \widehat{u}$
Δu	$- \xi ^2 \widehat{u}$

Hence (5.16) becomes

$$\begin{cases} \partial_t \widehat{u}(\xi, t) + |\xi|^2 \widehat{u}(\xi, t) = 0, \\ \widehat{u}(\xi, 0) = \widehat{u}_0(\xi). \end{cases} \quad (5.18)$$

Solving this ODE gives

$$\widehat{u}(\xi, t) = e^{-t|\xi|^2} \widehat{u}_0(\xi). \quad (5.19)$$

Applying $\mathcal{F}^{-1}$ and using (5.15),

$$u(x, t) = (\Phi_H(\cdot, t) * u_0)(x), \quad (5.20)$$

where the convolution is in the x -variable and

$$\Phi_H(x, t) := (2\pi)^{-n/2} (e^{-t|\xi|^2})^\vee(x), \quad t > 0. \quad (5.21)$$

The source calls Φ_H “the fundamental solution to the heat equation.”

Exercise 5.4 (Explicit heat kernel and normalization). Show that, for $t > 0$,

$$\Phi_H(x, t) = (4\pi t)^{-n/2} e^{-|x|^2/(4t)}, \quad (5.22)$$

$$\int_{\mathbb{R}^n} \Phi_H(x, t) dx = 1. \quad (5.23)$$

Hint: compute the one-dimensional Gaussian integral and use iterated integration.

Next. More rigorous analysis of the Fourier transform.